Oracle

Retrained on Homology-Restricted MegaScale

Zero-shot

Supervised, not trained on Megascale

Rectified gain @ k=96 (kcal/mol)

Predicted most stabilizing per domain

0
1
10

6.50 6.49 13.25 16.65
3.49 3.74 9.00 10.17
3.69 3.81 9.51 11.17
1.99 2.97 6.22 8.19
2.70 2.70 8.46 9.18
2.89 3.26 8.79 9.16
2.23 3.15 2.67 3.55
2.81 4.37 3.55 3.30
0.48 0.25 0.90 0.43
2.60 1.66 6.74 5.45
2.05 2.22 6.36 6.54

12.97 17.18 26.88 38.97
12.10 17.03 28.09 38.78
11.54 16.31 27.70 40.45
10.02 17.91 25.29 42.78
11.24 15.17 29.63 39.96
11.46 14.82 30.72 39.00
20.73 30.67 13.91 30.70
23.24 32.18 16.21 24.36
21.85 27.85 21.30 25.00
19.48 10.91 33.13 32.42
11.97 17.06 28.00 38.97

Top 96 per domain

ESM-MSR (p0) WT head

ESM-MSR (p0) combined

MutateEverything st1

ThermoMPNN

SPURS

ESM3-small-open unmasked

ProteinMPNN

ProSST

GeoDDG-3D

Rosetta

Δ hydrophobicity ≤ 0 & relative ASA ≤ 0.5

Δ hydrophobicity > 0 & relative ASA ≤ 0.5

Δ hydrophobicity ≤ 0 & relative ASA > 0.5

Δ hydrophobicity > 0 & relative ASA > 0.5

Top 96 per Homology-Restricted MegaScale test domain (33 domains with the feature, 35,295 single mutants); gain = sum of max(ddG, 0) over the picks predicted stabilizing (> 0); boxes: interquartile range with darker colored median line; dots: domains; dark bars & annotations: mean over domains; dashed lines: oracle mean.